

Day 35 - Lagrangian Examples

Me on my way to learn Lagrangian formalism



Me after learning the Lagrangian formalism



Announcements

- Last "Class" Week
- Homework 8 due Friday, April 18
- Next Week: Project Work and Discussion
- Final Project Due April 28th (no later than 11:59 pm)
- **No Final Exam**

Seminars This Week

TUESDAY, April 15, 2025

- **Theory Seminar**, 11:00am., FRIB 1200, **Speaker: Fnu Aaina Thapa, LLNL**, Title: “Deducing neutron capture on short-lived nuclei”
- **High Energy Physics Seminar**, 1:00 pm, 1400 BPS, **Speaker: Fei Yao, Brookhaven National Lab**, Title: Extracting Meson Distribution Amplitudes from Nonlocal Euclidean Correlations at Next-to-Next-to-Leading Order

WEDNESDAY, April 16, 2025

- **Astronomy Seminar**, 1:30 pm, 1400 BPS, **Undergraduate Thesis Talks** will be given the next two weeks
- **PER Seminar**, 3:00 pm., BPS 1400, **Speaker: Rebeckah Fussell, Cornell University**, Title: Comparing approaches to using large language models in science education research
- **FRIB Nuclear Science Seminar**, 3:30pm., FRIB 1300 Auditorium, **Speaker: Professor Alex Brown (FRIB)**, Title: Nuclear Science Advances at the MSU Cyclotron, NSCL, FRIB, ...

THURSDAY, April 17, 2025

- **Colloquium**, 3:30 pm, 1415 BPS, **Speaker: Jessie Christiansen, Caltech/IPAC**, Title: From Kepler to the Habitable Worlds Observatory: The Emerging Picture of Planet Populations
- **Astronomical Horizons Public Lecture Series**, 7:30 pm, Skye Theater, Abrams Planetarium, **Speaker: Marcel Yanez Reyes**, Title: From Event Horizons to Particle Collisions: The Geometry of the Extreme.

Stand Up for Higher Education

- Graduate Employee Union
- Union of Nontenure Track Faculty
- Union of Tenure System Faculty

Thursday, April 17th at 3pm

Please make time to show up!

www.dayofactionforhighered.org

STAND UP FOR HIGHER ED RALLY

**APRIL 17
3 PM**

FRONT OF THE HANNAH BUILDING

**SHOW YOUR SUPPORT
WEAR RED FOR ED
ANYWHERE YOU ARE**



Reminders

We used the Lagrangian formalism to derive the equations of motion for a plane pendulum. We chose the x and y coordinates.

$$T(\dot{x}, \dot{y}) = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) \quad V(y) = mgy$$

$$\mathcal{L} = T - V = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) - mgy$$

This gave us the following derivatives for the Lagrangian:

$$\frac{\partial \mathcal{L}}{\partial x} = 0 \quad \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) = \frac{d}{dt} (m\dot{x}) = 0$$

$$\frac{\partial \mathcal{L}}{\partial y} = -mg \quad \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{y}} \right) = -m\ddot{y}$$

We made a mistake by not including the constraint

We made a mistake by not including the constraint $x^2 + y^2 = L^2$ in our Lagrangian.

We can change variables to r and ϕ .

$$x = r \cos(\phi) \quad y = r \sin(\phi)$$

$$T(\dot{x}, \dot{y}) = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) = \frac{1}{2}m \left(r^2 \dot{\phi}^2 + 2r\dot{r}\dot{\phi} + \dot{r}^2 \right) = T(r, \dot{r}, \phi, \dot{\phi})$$

$$V(y) = mgy = mgr \sin(\phi) = V(r, \phi)$$

Now we include the constraint $r = L$, so that $\dot{r} = 0$.

$$T(\phi, \dot{\phi}) = \frac{1}{2}mL^2\dot{\phi}^2 \quad V(\phi) = mgL \cos(\phi)$$

$$\mathcal{L} = \frac{1}{2}mL^2\dot{\phi}^2 - mgL \cos(\phi)$$

Clicker Question 35-1

For the plane pendulum, we changed the Lagrangian from Cartesian coordinates to plane polar coordinates. In Cartesian, we found the Lagrangian depended on $y, \dot{x}, \dot{y}$. In polar, it only depended on ϕ and $\dot{\phi}$.

$$\mathcal{L}(x, y, \dot{y}) \longrightarrow \mathcal{L}(\phi, \dot{\phi})$$

What does that tell you about the dimensions of the system? The system is:

1. in 3D space, so it's 3D.
2. described by two spatial dimensions (x, y) , so it's 2D.
3. described by one spatial dimension (ϕ) , so it's 1D.

Clicker Question 35-2

With $\mathcal{L} = \frac{1}{2}mL^2\dot{\phi}^2 - mgL \cos(\phi)$, we can find the equations of motion.

$$\frac{\partial \mathcal{L}}{\partial \phi} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}} \right) = 0$$

Which of the following equations of motion is correct?

1. $\ddot{\phi} = -\frac{g}{L} \sin(\phi)$
2. $\ddot{\phi} = -\frac{g}{L} \cos(\phi)$
3. $\ddot{\phi} = -\sqrt{\frac{g}{L} \sin(\phi)}$
4. $\ddot{\phi} = -\sqrt{\frac{g}{L} \cos(\phi)}$
5. None of these

Clicker Question 35-3

For the Atwood's machine, M is connected to m by a string of length l . Each mass has a length of string extended as measured from the center of the pulley (R) of y_1 and y_2 , respectively. The string wraps around half the pulley.

Which of the following represents the equation of constraint for the system?

1. $y_1 + y_2 = l - R\phi$
2. $y_1 - y_2 = l + R\phi$
3. $y_1 + y_2 = l - \pi R$
4. $y_1 - y_2 = l + \pi R$
5. None of these

Take the time derivative of the constraint equation. What do you notice?

Clicker Question 35-4

With a Lagrangian of the form $\mathcal{L} = \frac{1}{2}(M + m)\dot{y}_1^2 - (M - m)gy_1$, we can find the **generalized forces** and **generalized momenta**.

$$F_{y_1} = \frac{\partial \mathcal{L}}{\partial y_1} = -\frac{\partial V}{\partial y_1} \quad p_{y_1} = \frac{\partial \mathcal{L}}{\partial \dot{y}_1} = \frac{\partial T}{\partial \dot{y}_1}$$

What are F_{y_1} and p_{y_1} for the Atwood's machine?

1. $F_{y_1} = -mg$ and $p_{y_1} = m\dot{y}_1$
2. $F_{y_1} = -Mgy_1$ and $p_{y_1} = M\dot{y}_1$
3. $F_{y_1} = -(M - m)g$ and $p_{y_1} = (M + m)\dot{y}_1$
4. $F_{y_1} = -(M + m)g$ and $p_{y_1} = (M - m)\dot{y}_1$
5. None of these

Clicker Question 35-5

Now, we allow the pulley (mass, M_p) to rotate. The Lagrangian is given by:

$$\mathcal{L} = \frac{1}{2}(M + m)\dot{y}_1^2 + \frac{1}{2}I\dot{\phi}^2 - (M - m)gy_1$$

Where I is the moment of inertia of the pulley. What is the moment of inertia of the pulley?

1. $I = \frac{1}{2}M_pR^2$
2. $I = \frac{1}{3}M_pR^2$
3. $I = M_pR^2$
4. $I = \frac{1}{4}M_pR^2$
5. None of these

Clicker Question 35-6

The rope moves without slipping on the pulley. A rotation of $Rd\phi$ corresponds to a displacement of dy_1 for the first mass, M . What is the **new** equation of constraint for the system?

1. $y_1 + y_2 = l - R\phi$

2. $dy_1 = Rd\phi$

3. $y_1 = R\phi$

4. $\dot{y}_1 = R\dot{\phi}$

5. More than one of these